

Databases

Laboratory work 1: ERD Diagram

Airports	
airport_id	integer
airport_name	varchar NN
country	varchar NN
state	varchar
city	varchar NN
created_at	timestamp
updated_at	timestamp

Airlines	
airline_id	integer
airline_code	varchar NN
name	varchar NN
country	varchar NN
created_at	timestamp
updated_at	timestamp

Passengers	
passenger_id	integer
first_name	varchar NN
last_name	varchar NN
gender	varchar
date_of_birth	date NN
country_of_citizenship	varchar NN
country_of_residence	varchar NN
passport_number	varchar NN
created_at	timestamp
updated_at	timestamp

Flights	
flight_id	integer
airline_id	integer NN
departure_airport_id	integer NN
arrival_airport_id	integer NN
departing_gate	varchar
arriving_gate	varchar
scheduled_departure_time	timestamp NN
scheduled_arrival_time	timestamp NN
actual_departure_time	timestamp
actual_arrival_time	timestamp
created_at	timestamp
updated_at	timestamp

Security Checks	
security_check_id	integer
passenger_id	integer NN
check_results	text NN
created_at	timestamp
updated_at	timestamp

Bookings	
booking_id	integer
flight_id	integer NN
passenger_id	integer NN
status	varchar NN
booking_platform	varchar NN
ticket_price	decimal(10,2) NN
created_at	timestamp
updated_at	timestamp

Booking_Changes	
change_id	integer
booking_id	integer NN
change_details	text NN
created_at	timestamp

Boarding Passes	
boarding_pass_id	integer
booking_id	integer NN
seat	varchar NN
boarding_time	timestamp NN
created_at	timestamp
updated_at	timestamp

Baggage	
baggage_id	integer
booking_id	integer NN
weight_kg	decimal(5,2) NN
created_at	timestamp
updated_at	timestamp

Baggage Checks	
baggage_checking_id	integer
booking_id	integer NN
passenger_id	integer NN
check_results	text NN
created_at	timestamp
updated_at	timestamp

International Airport Database — ERD Description

- The International Airport Database is designed to manage airport operations, including airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks, security checks, and booking changes.
- The main entities in the database are Airport, Airline, Flight, Passenger, Booking, BoardingPass, Baggage, BaggageCheck, SecurityCheck, and BookingFlightChange.

Airport stores information about airports, including airport name, country, state, city, and timestamps.

Airline stores information about airline companies, including airline code, name, country, and timestamps.

Flight stores scheduled and actual flight information. Each flight belongs to one airline and has one departure airport and one arrival airport. An airport can be used by many flights as a departure or arrival airport.

Passenger stores passenger personal and passport information.

Booking connects passengers with flights. One passenger can have many bookings, and one flight can have many bookings. Therefore, Booking represents the relationship between passengers and flights.

BoardingPass stores boarding pass information such as seat and boarding time and is associated with a booking.

Baggage stores information about baggage registered during a booking. One booking can have multiple baggage records.

BaggageCheck stores the results of baggage checks and is associated with both a booking and a passenger.

SecurityCheck stores information about passenger security checks.

BookingFlightChange stores information about changes made to a flight booking. One booking can have multiple changes.

Data Dictionary & Entity Details

•**Airports:** airport_id (PK, INT, NOT NULL), airport_name (VARCHAR, NOT NULL), country (VARCHAR, NOT NULL), state (VARCHAR), city (VARCHAR, NOT NULL),
•created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**Airlines:** airline_id (PK, INT, NOT NULL), airline_code (VARCHAR, UNIQUE, NOT NULL), name (VARCHAR, NOT NULL), country (VARCHAR, NOT NULL),
•created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**Flights:** flight_id (PK, INT, NOT NULL), airline_id (FK -> Airlines, INT, NOT NULL), departure_airport_id (FK -> Airports, INT, NOT NULL), arrival_airport_id (FK -> Airports, INT, NOT NULL),
•departing_gate (VARCHAR), arriving_gate (VARCHAR), scheduled_departure_time (TIMESTAMP, NOT NULL), scheduled_arrival_time (TIMESTAMP, NOT NULL),
• actual_departure_time (TIMESTAMP), actual_arrival_time (TIMESTAMP), created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**Passengers:** passenger_id (PK, INT, NOT NULL), first_name (VARCHAR, NOT NULL), last_name (VARCHAR, NOT NULL), gender (VARCHAR), date_of_birth (DATE, NOT NULL),
•country_of_citizenship (VARCHAR, NOT NULL), country_of_residence (VARCHAR, NOT NULL), passport_number (VARCHAR, UNIQUE, NOT NULL),
•created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**Bookings:** booking_id (PK, INT, NOT NULL), flight_id (FK -> Flights, INT, NOT NULL), passenger_id (FK -> Passengers, INT, NOT NULL), status (VARCHAR, NOT NULL),
booking_platform (VARCHAR, NOT NULL), ticket_price (DECIMAL, NOT NULL), created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**BookingFlightChanges:** change_id (PK, INT, NOT NULL), booking_id (FK -> Bookings, INT, NOT NULL), change_details (TEXT, NOT NULL), created_at (TIMESTAMP).

•**BoardingPasses:** boarding_pass_id (PK, INT, NOT NULL), booking_id (FK -> Bookings, INT, UNIQUE, NOT NULL), seat (VARCHAR, NOT NULL), boarding_time (TIMESTAMP, NOT NULL),
•created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**Baggage:** baggage_id (PK, INT, NOT NULL), booking_id (FK -> Bookings, INT, NOT NULL), weight_kg (DECIMAL, NOT NULL), created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**BaggageChecks:** baggage_checking_id (PK, INT, NOT NULL), booking_id (FK -> Bookings, INT, NOT NULL), passenger_id (FK -> Passengers, INT, NOT NULL),
• check_results (TEXT, NOT NULL), created_at (TIMESTAMP), updated_at (TIMESTAMP).

•**SecurityChecks:** security_check_id (PK, INT, NOT NULL), passenger_id (FK -> Passengers, INT, NOT NULL), check_results (TEXT, NOT NULL),
•created_at (TIMESTAMP), updated_at (TIMESTAMP).

Main Relationships

- One Airline can have many Flights (1).
- One Airport can have many departing Flights (1).
- One Airport can have many arriving Flights (1).
- One Passenger can have many Bookings (1).
- One Flight can have many Bookings (1).
- One Booking can have zero or one Boarding Pass (1:0..1).
- One Booking can have many Baggage records (1).
- One Booking can have many Baggage Checks (1).
- One Passenger can have many Baggage Checks (1).
- One Passenger can have many Security Checks (1).
- One Booking can have many Booking Flight Changes (1).

Notation

PK () means Primary Key and uniquely identifies a record.

FK () means Foreign Key and references a primary key in another entity.

NOT NULL (NN) means that an attribute is required.

UNIQUE () means that an attribute cannot contain duplicate values.